

PYTHON FULL STACK TRAINING: DICTIONARY LOGIC & INTERVIEW PREP

Topic: Key-Value Pairs, JSON-like Structures, and Data Mapping

--- PART 1: DICTIONARY BASICS & CREATION ---

1. Create a dictionary representing a 'Laptop' with keys: brand, model, and price.
2. Access the value of the 'model' key using square brackets.
3. Access the value of a key that doesn't exist using `.get()` and explain why it's safer than `[]`.
4. Create an empty dictionary using both `{}` and the `dict()` constructor.
5. Add a new key-value pair 'processor': 'i7' to an existing dictionary.
6. Update the 'price' of the laptop to a new value.
7. Use the `len()` function to find how many key-value pairs are in a dictionary.
8. Create a dictionary where the keys are numbers 1 to 5 and the values are their squares.
9. Check if a specific key exists in a dictionary using the 'in' operator.
10. Delete a key-value pair using the 'del' keyword and handle the case if the key is missing.

--- PART 2: METHODS & MANIPULATION ---

11. Use the `pop()` method to remove a key and store its value in a variable.
12. Use `popitem()` to remove the last inserted item and explain its behavior in Python 3.7+.
13. Use the `keys()` method to print all the keys in a dictionary.
14. Use the `values()` method to print all the values in a dictionary.
15. Use the `items()` method to iterate through a dictionary and print "Key: Value" for each pair.
16. Merge two dictionaries: `{'a': 1, 'b': 2}` and `{'c': 3, 'd': 4}` using the `update()` method.
17. Clear all items from a dictionary using the `clear()` method.
18. Use the `setdefault()` method to add a key 'country' with value 'India' only if it doesn't exist.
19. Create a shallow copy of a dictionary and show that modifying the copy doesn't change the original.
20. Create a dictionary from two lists: one for keys and one for values using the `zip()` function.

--- PART 3: NESTED DICTIONARIES (API STYLE) ---

21. Create a nested dictionary called 'Employees' containing data for three different people.
22. Access a value inside a nested dictionary (e.g., Employees['emp1']['salary']).
23. Update a value inside a nested dictionary.
24. Add a new nested dictionary (a new employee) to the existing 'Employees' structure.
25. Write a loop to print only the names of all employees from the nested 'Employees' dictionary.
26. Create a dictionary where a key points to a list of values (e.g., 'hobbies': ['coding', 'reading']).
27. Append a new hobby to the list inside that dictionary.
28. Given a dictionary of students and their marks (a list), calculate the average marks for one student.
29. Flatten a simple nested dictionary (convert {'a': {'b': 1}} to {'a', 'b': 1}).
30. Represent a JSON response from a weather API as a nested dictionary and extract the 'temperature'.

--- PART 4: COMPREHENSIONS & LOGIC ---

31. [Comprehension] Create a dictionary of even numbers between 1-10 as keys and their cubes as values.
32. [Comprehension] Given a dictionary, create a new one with only items where the value is > 100.
33. [Comprehension] Swap keys and values in a dictionary (Reverse Mapping).
34. Sort a dictionary by its keys in alphabetical order.
35. Sort a dictionary by its values in ascending order.
36. Find the key with the maximum value in a dictionary of product prices.
37. Count the frequency of each character in a string "Python Trainer" using a dictionary.
38. Combine two dictionaries by adding values for common keys.
39. Convert a dictionary into a list of tuples.
40. Check if all values in a dictionary are the same.

--- PART 5: INDUSTRIAL & INTERVIEW SCENARIOS ---

41. [JSON] Use the 'json' module to convert a dictionary into a JSON string.

42. [Data Cleaning] Remove all keys from a dictionary that have None or empty string values.
43. [Security] Explain why a list cannot be used as a dictionary key, but a tuple can.
44. [Logic] Write a program to find the sum of all values in a numeric dictionary.
45. [Simulation] Use a dictionary to simulate a simple "Switch-Case" statement logic.
46. [Efficiency] Compare the time it takes to find a value in a list of 10,000 items vs. a dictionary.
47. [Functionality] Use `**kwargs` in a function to accept arbitrary keyword arguments as a dictionary.
48. [Mapping] Create a dictionary to map Roman numerals to Integers (e.g., 'I': 1, 'V': 5).
49. [Logic] Given a list of words, group them by their starting letter using a dictionary.
50. [Interview Question] Explain the difference between '==' and 'is' when comparing two identical dictionaries.